

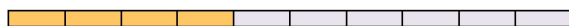
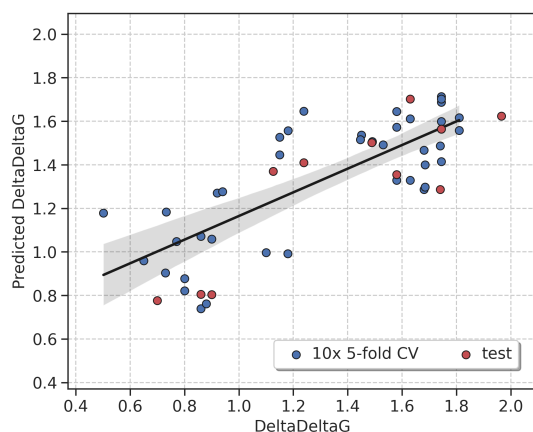


ROBERT v 2.0.1 2025/04/16 20:03:39

How to cite: Dalmau, D.; Alegre Requena, J. V. WIREs Comput Mol Sci. 2024, 14, e1733.**Section A. ROBERT Score***This score is designed to evaluate the models using different metrics.***No PFI (standard descriptor filter) · Score 4**

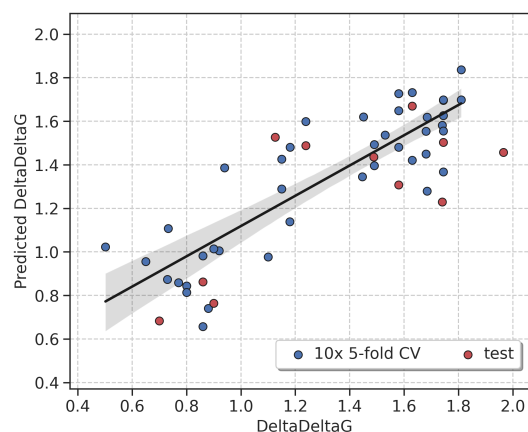
Model = GB · CV (train+valid.):Test = 79:21

Points(train+validation):descriptors = 42:18

**WEAK**10x 5-fold CV : $R^2 = 0.51$, MAE = 0.22, RMSE = 0.28Test : $R^2 = 0.74$, MAE = 0.18, RMSE = 0.22**PFI (only important descriptors) · Score 3**

Model = GB · CV (train+valid.):Test = 79:21

Points(train+validation):descriptors = 42:5

**VERY WEAK**10x 5-fold CV : $R^2 = 0.68$, MAE = 0.18, RMSE = 0.23Test : $R^2 = 0.55$, MAE = 0.22, RMSE = 0.29**Severe warnings**☐ No severe warnings detected**Moderate warnings**☐ No moderate warnings detected**Overall assessment**☒ The model is unreliable**Severe warnings**☐ No severe warnings detected**Moderate warnings**☐ No moderate warnings detected**Overall assessment**☒ The model is unreliable



Section B. Advanced Score Analysis

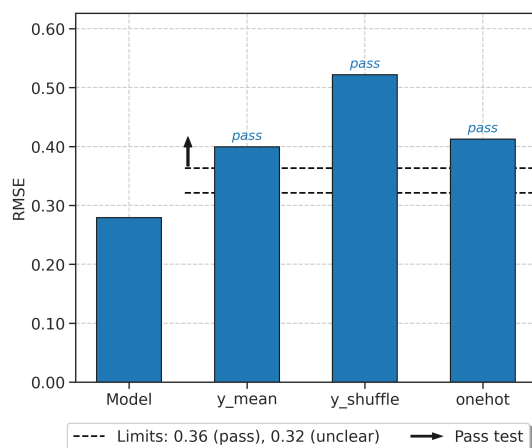
This section explains each component that comprises the ROBERT score. [More details here.](#)

1. Model vs "flawed" models (0 / 0)

The model predicts right for the right reasons.

· Scoring from -6 to 0 ·

Pass: 0, Unclear: -1, Fail: -2.

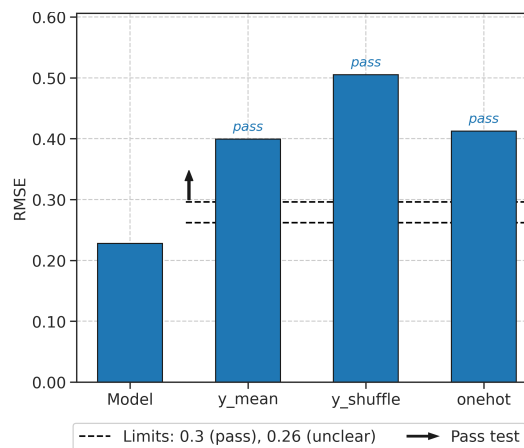


1. Model vs "flawed" models (0 / 0)

The model predicts right for the right reasons.

· Scoring from -6 to 0 ·

Pass: 0, Unclear: -1, Fail: -2.



2. CV predictions of the model (0 / 2 ☐)

Scaled RMSE (10x 5-fold CV) = 18.67%.

R^2 (10x 5-fold CV) = 0.51.

· Scoring from 0 to 2 ·

Scaled RMSE $\leq$ 10%: +2, Scaled RMSE $\leq$ 20%: +1.

$R^2 < 0.5$: -2, $R^2 < 0.7$: -1

2. CV predictions of the model (0 / 2 ☐)

Scaled RMSE (10x 5-fold CV) = 15.33%.

R^2 (10x 5-fold CV) = 0.68.

· Scoring from 0 to 2 ·

Scaled RMSE $\leq$ 10%: +2, Scaled RMSE $\leq$ 20%: +1.

$R^2 < 0.5$: -2, $R^2 < 0.7$: -1

3. Predictive ability & overfitting

3a. Predictions test set (1 / 2 ☐)

Scaled RMSE (test set) = 14.67%.

R^2 (test set) = 0.74.

· Scoring from 0 to 2 ·

Scaled RMSE $\leq$ 10%: +2, Scaled RMSE $\leq$ 20%: +1.

$R^2 < 0.5$: -2, $R^2 < 0.7$: -1

3. Predictive ability & overfitting

3a. Predictions test set (0 / 2 ☐)

Scaled RMSE (test set) = 19.33%.

R^2 (test set) = 0.55.

· Scoring from 0 to 2 ·

Scaled RMSE $\leq$ 10%: +2, Scaled RMSE $\leq$ 20%: +1.

$R^2 < 0.5$: -2, $R^2 < 0.7$: -1

3b. Prediction accuracy test vs CV (2 / 2 ☐)

Relative differences in values from sections 2 and 3a.

RMSE in test is 0.79*scaled RMSE (CV).

· Scoring from 0 to 2 ·

Scaled RMSE (test) $\leq$ 1.25*scaled RMSE (CV): +2.

Scaled RMSE (test) $\leq$ 1.50*scaled RMSE (CV): +1.

3b. Prediction accuracy test vs CV (1 / 2 ☐)

Relative differences in values from sections 2 and 3a.

RMSE in test is 1.26*scaled RMSE (CV).

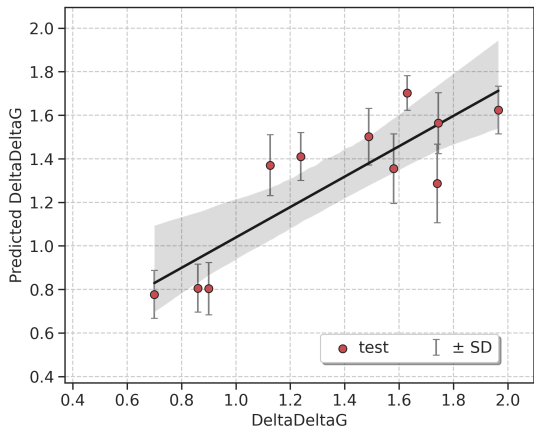
· Scoring from 0 to 2 ·

Scaled RMSE (test) $\leq$ 1.25*scaled RMSE (CV): +2.

Scaled RMSE (test) $\leq$ 1.50*scaled RMSE (CV): +1.

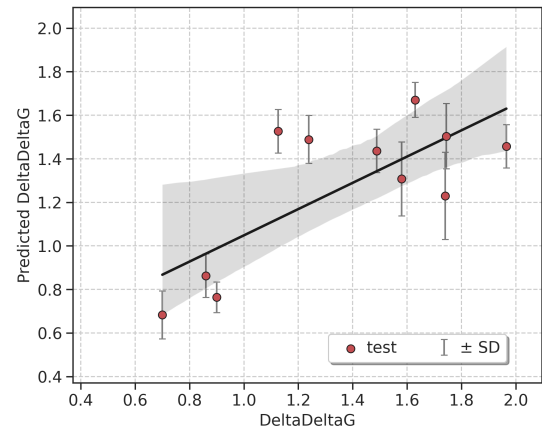
3c. Avg. standard deviation (SD) (1 / 2)

Moderate variation, 4*SD = 0.5 (35% y-range).
· Scoring from 0 to 2 ·
4*SD ≤ 25% y-range: +2, 4*SD ≤ 50% y-range: +1.



3c. Avg. standard deviation (SD) (1 / 2)

Moderate variation, 4*SD = 0.5 (32% y-range).
· Scoring from 0 to 2 ·
4*SD ≤ 25% y-range: +2, 4*SD ≤ 50% y-range: +1.



3d. Extrapolation (sorted CV) (0 / 2)

Scaled RMSEs across 5-fold CV:
[32.0%, 24.67%, 12.67%, 24.0%, 26.0%]
· Scoring from 0 to 2 ·
Every two folds with RMSEs ≤ 1.25*min RMSE: +1.

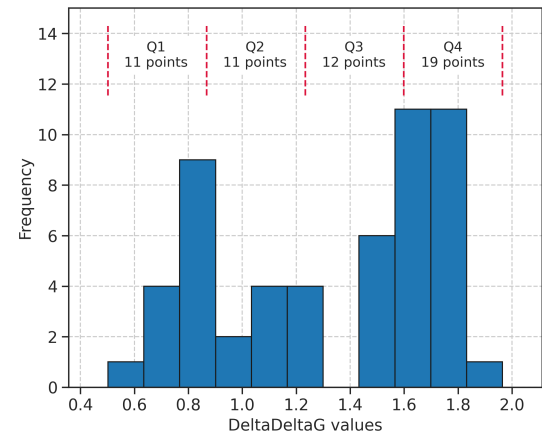
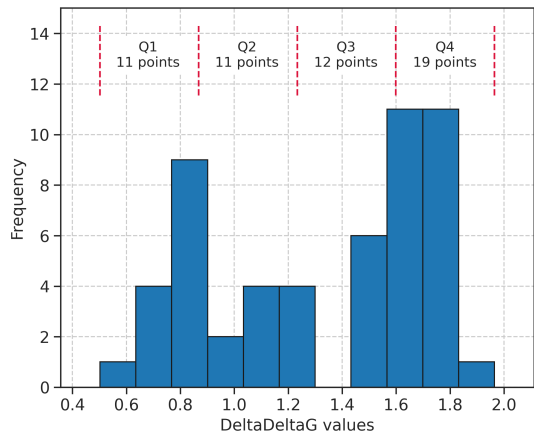
3d. Extrapolation (sorted CV) (1 / 2)

Scaled RMSEs across 5-fold CV:
[28.67%, 20.67%, 12.0%, 14.0%, 13.33%]
· Scoring from 0 to 2 ·
Every two folds with RMSEs ≤ 1.25*min RMSE: +1.



Section C. Distribution of y Values

This section shows the distribution of y values within the training and validation sets.



y distribution analysis

o Your data seems quite uniform

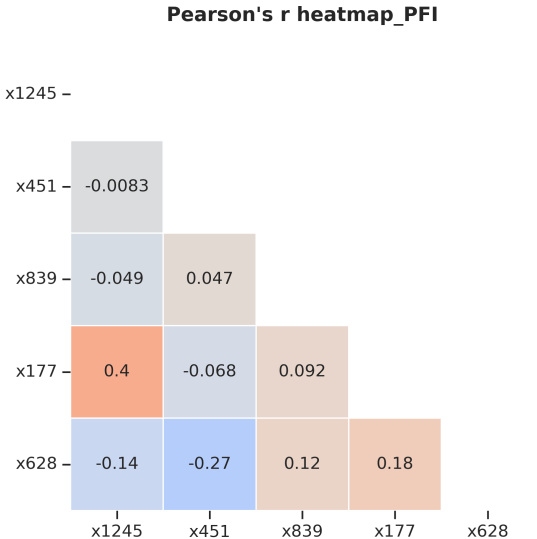
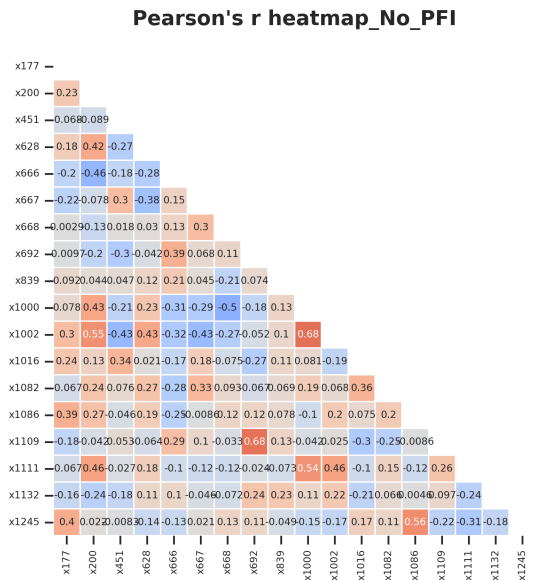
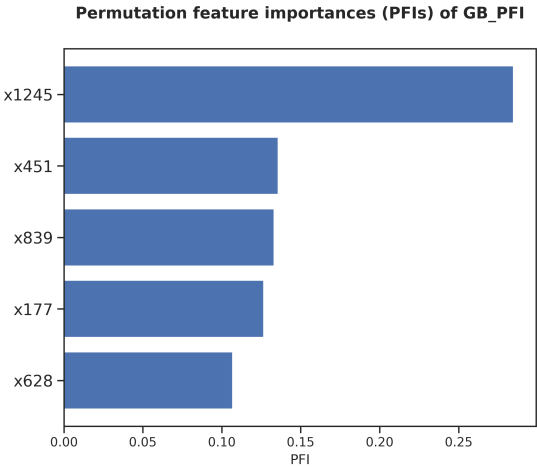
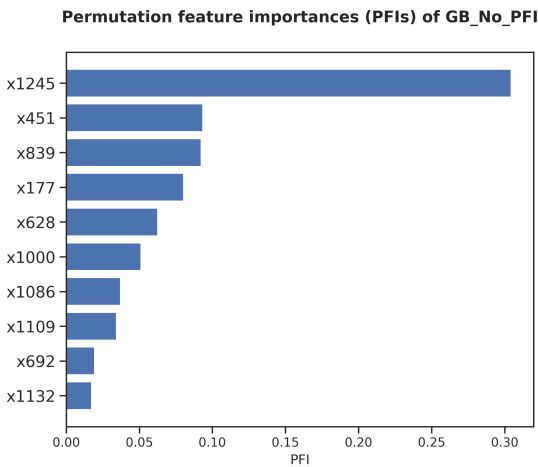
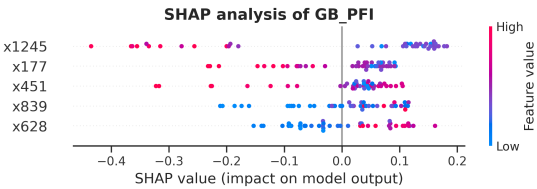
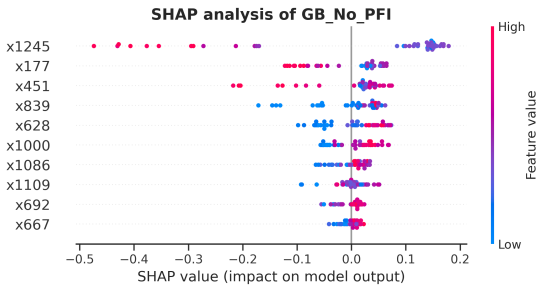
y distribution analysis

o Your data seems quite uniform



Section D. Feature Importances

This section presents feature importances measured using the validation set.



Correlation analysis

o Correlations between variables are acceptable

Correlation analysis

o Correlations between variables are acceptable



Section E. Outlier Analysis

This section detects outliers using the standard deviation (SD) of errors from the training set.

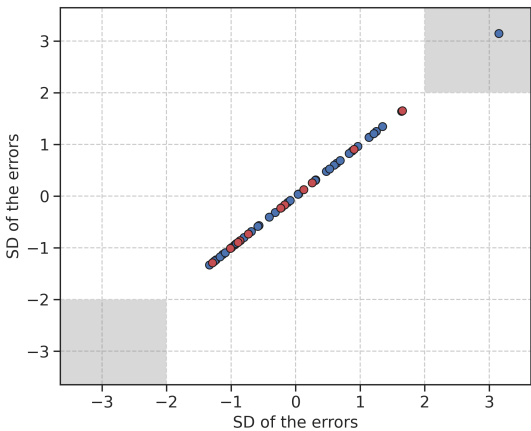
No PFI (standard descriptor filter):

Outliers (max. 10 shown)

Train: 1 outliers out of 42 datapoints (2.4%)

- L6_P26 (3.1 SDs)

Test: 0 outliers out of 11 datapoints (0.0%)



PFI (only important descriptors):

Outliers (max. 10 shown)

Train: 2 outliers out of 42 datapoints (4.8%)

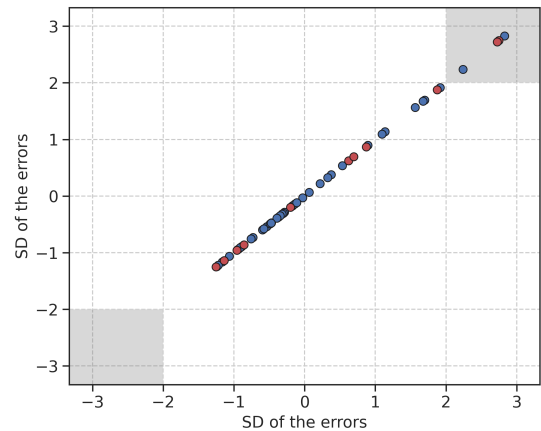
- L11_P0 (2.2 SDs)

- L6_P26 (2.8 SDs)

Test: 2 outliers out of 11 datapoints (18.2%)

- L3_P0 (2.7 SDs)

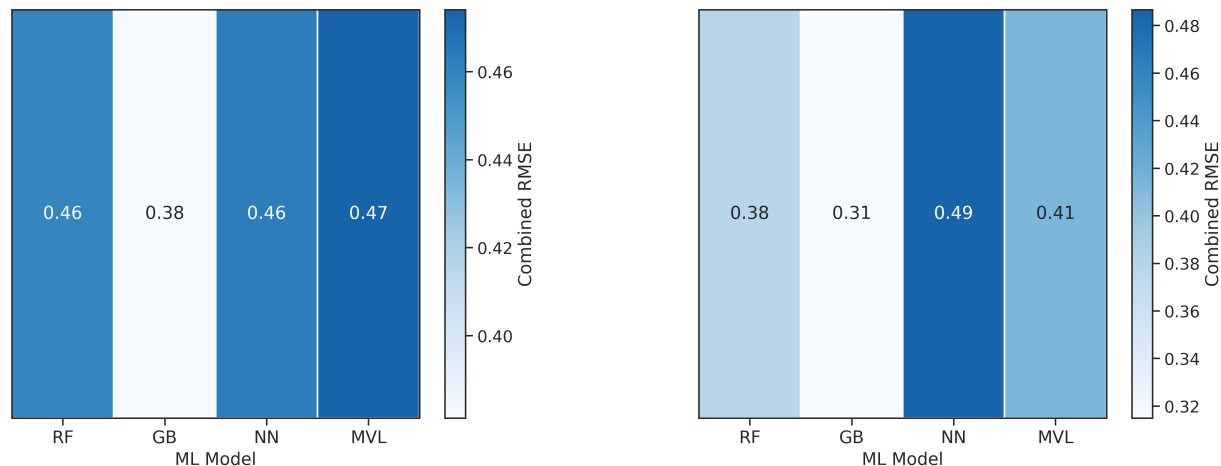
- L6_P9 (2.7 SDs)





Section F. Model Screening

This section compares different combinations of hyperoptimized algorithms and partition sizes. The combined error is calculated as the product of the training error, validation error, and cross-validation error.



Section G. Reproducibility

This section provides all the instructions to reproduce the results presented.

1. Download these files (the authors should have uploaded the files as supporting information!):

- CSV database (Lig_Sub_selected.csv)
- External test set (External_Test_Lig_Sub_selected.csv)

2. Install and adjust the versions of the following Python modules:

- Install ROBERT and its dependencies: `conda install -y -c conda-forge robert`
- Adjust ROBERT version: `pip install robert==2.0.1`
- Install scikit-learn-intelex: `pip install scikit-learn-intelex==2025.2.0`

(if scikit-learn-intelex is not installed, slightly different results might be obtained)

3. Run ROBERT using this command line in the folder with the CSV databases:

```
python -m robert --names "Structure" --y "DeltaDeltaG" --csv_name "Lig_Sub_selected.csv" --csv_test  
"External_Test_Lig_Sub_selected.csv"
```

4. Execution time, Python version and OS:

Originally run in Python 3.10.17 using Linux #1 SMP Mon Mar 10 11:32:40 UTC 2025

Total execution time: 484.32 seconds (the number of processors should be specified by the user)



Section H. Transparency

This section contains important parameters used in scikit-learn models and ROBERT.

1. Parameters of the scikit-learn models (same keywords as used in scikit-learn):

No PFI (standard descriptor filter):

sklearn model: GradientBoostingRegressor
 n_estimators: 39
 learning_rate: 0.2035490101894677
 max_depth: 7
 min_samples_split: 8
 min_samples_leaf: 3
 subsample: 0.754957408602135
 max_features: 0.6898847011075624
 validation_fraction: 0.10402150923749871
 min_weight_fraction_leaf: 0.04144700146086816
 ccp_alpha: 4.695476192547066e-05
 random_state: 0

PFI (only important descriptors):

sklearn model: GradientBoostingRegressor
 n_estimators: 39
 learning_rate: 0.2035490101894677
 max_depth: 7
 min_samples_split: 8
 min_samples_leaf: 3
 subsample: 0.754957408602135
 max_features: 0.6898847011075624
 validation_fraction: 0.10402150923749871
 min_weight_fraction_leaf: 0.04144700146086816
 ccp_alpha: 4.695476192547066e-05
 random_state: 0

2. ROBERT options, including prediction type (REG or CLAS), folds and repeats used for CV, etc:

No PFI (standard descriptor filter):

type: reg
 kfold: 5
 repeat_kfolds: 10
 seed: 0
 error_type: rmse

PFI (only important descriptors):

type: reg
 kfold: 5
 repeat_kfolds: 10
 seed: 0
 error_type: rmse



Section I. Abbreviations

Reference section for the abbreviations used.

ACC: accuracy	KN: k-nearest neighbors	REG: Regression
ADAB: AdaBoost	MAE: root-mean-square error	RF: random forest
CSV: comma separated values	MCC: Matthew's correl. coefficient	RMSE: root mean square error
CLAS: classification	ML: machine learning	RND: random
CV: cross-validation	MVL: multivariate lineal models	SHAP: Shapley additive explanations
F1 score: balanced F-score	NN: neural network	VR: voting regressor
GB: gradient boosting	PFI: permutation feature importance	
GP: gaussian process	R2: coefficient of determination	



Section J. New Predictions

Predictions of the external test set added with the csv_test option.

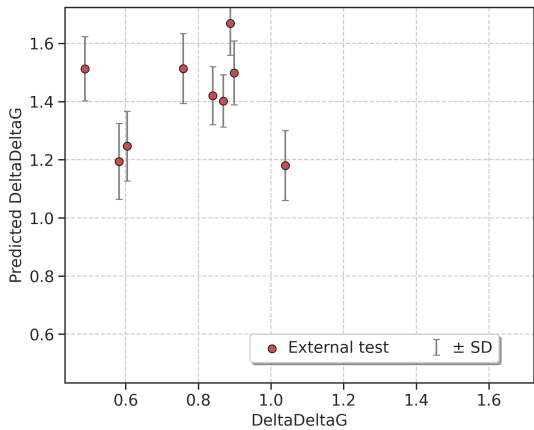
External test metrics

R2 = 0.0053, MAE = 0.63, RMSE = 0.67

External test predictions (sorted, max. 20 shown)

From /PREDICT/csv_test/...No_PFI.csv

Structure	DeltaDeltaG	DeltaDeltaG_pred ± sd
L7_P23	0.89	1.67 ± 0.11
L8_P23	0.76	1.51 ± 0.12
L15_P16	0.49	1.51 ± 0.11
L26_P23	0.9	1.5 ± 0.11
L29_P23	0.84	1.42 ± 0.1
L27_P23	0.87	1.4 ± 0.09
L16_P16	0.6	1.25 ± 0.12
L11_P16	0.58	1.19 ± 0.13
L14_P23	1.04	1.18 ± 0.12



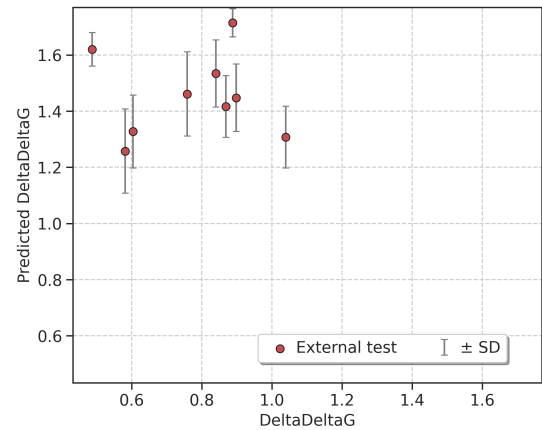
External test metrics

R2 = 0.00028, MAE = 0.68, RMSE = 0.71

External test predictions (sorted, max. 20 shown)

From /PREDICT/csv_test/..._PFI.csv

Structure	DeltaDeltaG	DeltaDeltaG_pred ± sd
L7_P23	0.89	1.71 ± 0.05
L15_P16	0.49	1.62 ± 0.06
L29_P23	0.84	1.53 ± 0.12
L8_P23	0.76	1.46 ± 0.15
L26_P23	0.9	1.45 ± 0.12
L27_P23	0.87	1.42 ± 0.11
L16_P16	0.6	1.33 ± 0.13
L14_P23	1.04	1.31 ± 0.11
L11_P16	0.58	1.26 ± 0.15



Miscellaneous

General tips to improve the models and instructions to predict new values.

Some general tips to improve the score

1. Adding meaningful datapoints might help to improve the model. Also, using a uniform population of datapoints across the whole range of y values usually helps to obtain reliable predictions across the whole range. More information about the range of y values used is available in Section C.
2. Adding meaningful descriptors or replacing/deleting the least useful descriptors used might help. Feature importances are gathered in Section D.

How to predict new values with these models?

1. Create a CSV database with the new points, including the necessary descriptors.
 2. Place the CSV file in the parent folder (i.e., where the module folders were created)
 3. Run the PREDICT module as 'python -m robert --predict --csv_test FILENAME.csv'.
 4. The predictions will be shown at the end of the resulting PDF report and will be stored in the last column of two CSV files called MODEL_SIZE_test(_No)_PFI.csv, which are in the PREDICT folder.
-